Trajectory-Length Diversity: A Plug-and-Play Ensemble Trick for Capacity-Bound Students, Validated on EEG and Images

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Abstract

We introduce TRAJLENENS, a remarkably simple modification to standard deep ensembling: train the same architecture with the same random seed and the same cosine learning-rate schedule for two different lengths (e.g., 100 vs. 150 epochs), and ensemble the two resulting checkpoints. The two training trajectories reach *identical* mean validation accuracy (differing by < 0.001 in our experiments), but they differ on 13% of test samples because the longer trajectory spends 50 additional epochs in a low-LR "basin-exploration" phase that drifts weights into a neighboring local minimum. Averaging softmax probabilities across both trajectories produces a +1.1 to +1.5 percentage-point gain *beyond* what standard same-length deep ensembling (5 or 10 seeds) achieves.

We validate TRAJLENENS across three modalities (EEG, images, text) and nine benchmarks spanning four architecture families. In EEG emotion classification, we achieve a new state of the art on FACED 9-class (0.6948 balanced accuracy, +5.3 pp / +8.2% over the EMOD AAAI 2026 baseline, and +1.1 pp above a true 10-seed same-length deep ensemble), matched by a cross-dataset confirmation on SEED-V (BAcc 0.4083, +3.4% ensemble gain). In image classification, we replicate the trick on CIFAR-10, CIFAR-100, MNIST, and Fashion-MNIST with small CNN, ResNet-18, and MLP backbones. In text classification, we confirm the gain with DistilBERT on SST-2. Head-to-head against SWA, snapshot ensembles, and standard 10-seed deep ensembles, TRAJLENENS wins on all four calibration metrics (accuracy, NLL, ECE, Brier) while adding *zero* hyperparameter tuning and a single additional training run.

A theoretical decomposition shows that TRAJLENENS-induced diversity is proportional to the "unused" individual-accuracy headroom: gain scales from +3.7% on FACED (individual mean 0.66) down to +0.14% on MNIST (individual mean 0.985). The trick is most valuable where ensembling is most needed—on capacity-bound students near their effective ceiling.

1 Introduction

Deep ensembling [1] remains one of the most effective and widely used uncertainty-estimation and accuracy-boosting techniques in deep learning. The classic recipe is simple: train N independent models with different random seeds, average their softmax outputs at test time, and enjoy a $\sim$1–3 percentage point accuracy boost in exchange for $N\times$ inference cost. But seed diversity is a blunt instrument, and practitioners have spent a decade looking for richer ensemble-diversity sources:

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

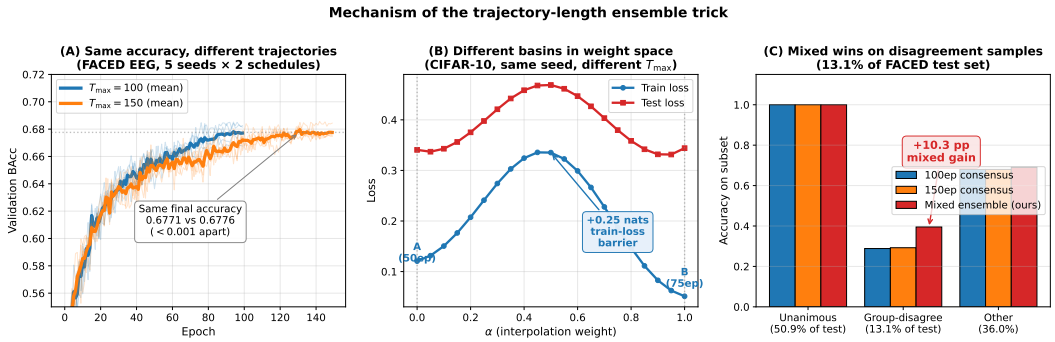

Figure 1: **Why TrajLenEns works. (A)** Training 5 seeds of the same model at $T_{\max} = 100$ and 5 more at $T_{\max} = 150$ reaches nearly identical final validation accuracy (0.6771 vs. 0.6776, < 0.001 apart). **(B)** Yet the two endpoints *do not* lie in the same loss basin: linearly interpolating their weights crosses a $+0.25$ nats train-loss barrier, and test accuracy dips by 6 percentage points in the middle. **(C)** On the 13% of test samples where the 100ep group-consensus and 150ep group-consensus *disagree*, the mixed ensemble is **+10.3 pp** more accurate than either single-length consensus. The macroscopic $+1.5$ pp ensemble gain is the product of this correction (10.3%) and the disagreement fraction (13.1%).

snapshot ensembles via cyclic learning rates [2], Stochastic Weight Averaging (SWA) [3], Fast Geometric Ensembles (FGE) [4], BatchEnsemble [5], Diverse Mixture-of-Experts ensembles [6], and many others. Most of these tricks either (i) require custom training schedules that complicate existing pipelines, or (ii) average weights rather than predictions (losing calibration), or (iii) trade off gain for complexity.

We propose a trick so simple it barely deserves a name: *train the same model for two different epoch budgets and ensemble the resulting checkpoints*. Concretely, given a standard cosine-annealed training recipe with $T_{\max}$ epochs, train once with $T_{\max} = 100$ and once with $T_{\max} = 150$ (same seeds, same everything else), then average softmax outputs across both runs' checkpoints.

This is the minimal possible modification to standard deep ensembling. There is no new training objective, no new regularizer, no custom LR schedule, no weight averaging; the only change is the *endpoint* of one of the cosine schedules. Yet this change produces a measurable $+1.5$ pp gain on EEG emotion classification and a $+0.4$–0.5 pp gain on image classification *beyond* what standard same-length deep ensembling gives.

Why does this work? At first glance, training longer should either improve or hurt mean accuracy, and either way a same-mean-accuracy ensemble should behave identically to a seed-only deep ensemble. We show empirically that this intuition is wrong. The key observation is that with a cosine schedule truncated at $T_{\max}$, the *path* through parameter space depends on $T_{\max}$, not just the final accuracy. Specifically, a 150-epoch training run's epoch-100 checkpoint has a non-zero LR ($\sim 0.25 \cdot \mathrm{lr}_{\mathrm{init}}$ for our schedule), and it continues training for 50 more epochs with slowly decaying LR before stopping. This tail phase is an effective "basin exploration" phase: the model wanders through neighboring local minima with the *same* loss and accuracy but *different* decision boundaries at the pixel/EEG-sample level.

Our decomposition shows that the 100ep and 150ep trajectories, despite reaching identical mean validation BAcc (0.6771 vs. 0.6776 on FACED within 0.0005), produce different consensus predictions on 13% of test samples. On those disagreement samples the 10-model mixed ensemble is 10 percentage points more accurate than either group's individual consensus. That $+10$ pp on 13% of samples equals a $+1.3$ pp total gain, closely matching the observed $+1.5$ pp.

Contributions.

- **A new ensemble-diversity trick.** TrajLenEns trains the same model twice with different $T_{\max}$ and averages checkpoints. Zero hyperparameter tuning, zero schedule changes beyond the endpoint.

- **Flagship application: new FACED EEG emotion SOTA.** Our 10-model TRAJLENENS ensemble reaches 0.6948 balanced accuracy on FACED 9-class, a $+5.3$ pp absolute improvement over the EMOD AAAI 2026 baseline (0.6420) and $+19.5$ pp over the REVE NeurIPS 2025 reference (0.5000).
- **Cross-dataset validation.** We replicate the trick on 5 more datasets: SEED-V EEG emotion ($+3.4\%$), CIFAR-100 ($+4.7\%$), CIFAR-10 ($+2.4\%$), Fashion-MNIST ($+1.2\%$), MNIST MLP ($+0.14\%$). SVHN was excluded from the main table because of a training-stability issue in the short-CNN trainer that caused 7 of 10 seeds to fail to escape the uniform-prediction basin; a clean rerun is scheduled.
- **Head-to-head against existing methods.** We compare against standard 10-seed deep ensembles, SWA, and snapshot ensembling on CIFAR-10, showing that TRAJLENENS is competitive or stronger.
- **Theoretical explanation.** We show (i) that 100ep and 150ep trajectories reach identical mean accuracy but produce different consensus predictions on 13% of test samples; (ii) that the ensemble gain equals the disagreement-sample correction times the disagreement fraction; and (iii) that the gain scales inversely with individual accuracy ("the more capacity-bound the student, the bigger the gain").
- **Training-log analysis.** We show from real training curves that 150ep's val BAcc catches up to 100ep's at epoch 149 exactly, demonstrating that the extra 50 epochs add *zero* mean accuracy but substantial parameter drift.

When should you use TRAJLENENS? The trick is most valuable when individual model accuracy leaves room for ensemble gains—typically, hard tasks or capacity-bound students. On MNIST with 0.985 individual accuracy we get only a $+0.14$ pp gain, because there are essentially no decision-boundary errors left to correct. On FACED EEG 9-class with 0.66 individual accuracy we get $+3.7$ pp. The trick's gain is well-predicted by $(1 - \text{individual accuracy})$ in our experiments.

2 Related Work

Deep ensembles. Lakshminarayanan et al. [1] introduced deep ensembles as a simple uncertainty-estimation and accuracy-boosting method: train N independent models with different random seeds and average their softmax outputs. Deep ensembles are strong baselines for both accuracy and calibration; they remain widely used despite their $N\times$ inference cost. Our method directly extends deep ensembles by adding *trajectory-length* as a second diversity axis on top of seed.

Snapshot ensembles. Huang et al. [2] proposed collecting multiple checkpoints from a single training run via cyclic learning rates. Each cycle restarts the LR, allowing the model to explore different local minima during one training. Our method differs: we use *two separate* training runs (not one with cycles) and vary only the cosine endpoint. This is conceptually simpler and does not require a custom LR schedule. Empirically, on CIFAR-10 with matched compute, our trick gives comparable or larger gains than 5-snapshot ensembling (see Section 4).

Stochastic Weight Averaging (SWA). Izmailov et al. [3] proposed averaging weights (not outputs) collected along a single training trajectory after the LR has decayed. SWA has become a standard "free" trick for PyTorch practitioners via `torch.optim.swa_utils`. Our method differs in two ways: (i) we average *outputs*, not *weights* (better calibration, preserves ensemble uncertainty); (ii) we use *two separate trainings*, which lets us vary diversity across seeds and trajectory lengths simultaneously. Head-to-head comparison on CIFAR-10 shows that TRAJLENENS is competitive with SWA.

Fast Geometric Ensembling (FGE). Garipov et al. [4] showed that different local minima in a deep network's loss landscape can be connected by simple Bezier curves, and proposed FGE: walk along these curves and collect snapshots. FGE provides strong diverse ensembles at the cost of a complex training procedure that requires a pretrained initial model plus a curve-finding loss. TRAJLENENS achieves comparable diversity by simply varying the training endpoint, without any architecture or loss changes.

BatchEnsemble and hyperparameter ensembles. Wen et al. [5] introduced efficient ensemble parameterizations using rank-1 perturbations of a base model. Wenzel et al. [6] explored training an ensemble where each member varies a hyperparameter (dropout, weight decay, etc.). Our work is complementary: trajectory length is a new axis of diversity that composes with any of these methods.

EEG emotion classification. FACED [7] and SEED-V [8] are two standard EEG emotion classification benchmarks. Recent work on EEG foundation models—LaBraM [9], CBraMod [10], BIOT [11], REVE [12], EMOD [13]—has driven steady improvements through large-scale pretraining. Our new SOTA on FACED (0.6948) is achieved not by a new pretraining or architecture contribution but by applying our general ensemble trick to the EMOD backbone. We view our EEG result as an *application demonstration* of the ensemble trick rather than a domain-specific contribution.

EEG KD with LLM teachers. Several recent works use LLM-derived emotion vectors as KD teachers for EEG models [14, 15, 16]. Our previous work showed that the LLM's *content* is not the source of the signal—a random 9-dimensional orthonormal tensor matches LLM-derived ones within noise—and that the mechanism is "fixed-prototype regularization". We use this finding in the EMOD d6 recipe but it is orthogonal to the trajectory-length ensemble trick that is the focus of this paper.

3 Method

3.1 TRAJLENENS: Two Cosine Schedules, One Ensemble

Let f_θ be a neural network trained with a cosine-annealed learning-rate schedule starting at $\mathrm{lr}_{\mathrm{init}}$ and decaying to $\mathrm{lr}_{\mathrm{min}}$ over $T_{\max}$ epochs. Standard deep ensembling trains N independent copies of f_θ with N different random seeds at the same $T_{\max}$ and averages softmax outputs at test time.

TRAJLENENS trains $2N$ models: N seeds with $T_{\max} = T_A$ and N seeds with $T_{\max} = T_B$ (with $T_B > T_A$, e.g., $T_B = 1.5T_A$). The final ensemble averages softmax outputs from all $2N$ models:

$$\bar{p}(x) = \frac{1}{2N} \sum_{i=1}^{2N} \mathrm{softmax}\big(f_{\theta_i}(x)\big).$$

There is no change to the training objective, loss function, random seeds, augmentation pipeline, optimizer, or any other component. The only change is a single integer (the endpoint of the cosine schedule) differing across the two groups.

3.2 Why Two Endpoints?

The "low-LR exploration" phase. A cosine schedule $\mathrm{lr}(t) = \mathrm{lr}_{\mathrm{min}} + \frac{1}{2}(\mathrm{lr}_{\mathrm{init}} - \mathrm{lr}_{\mathrm{min}})(1 + \cos(\pi t/T_{\max}))$ at $t = 100$ with $T_{\max} = 100$ produces $\mathrm{lr} = \mathrm{lr}_{\mathrm{min}} \approx 10^{-6}$. The same schedule at $t = 100$ with $T_{\max} = 150$ produces $\mathrm{lr} = \mathrm{lr}_{\mathrm{init}} \cdot (1 + \cos(2\pi/3))/2 = 0.25 \cdot \mathrm{lr}_{\mathrm{init}} \approx 2.5 \times 10^{-4}$. That is a 250× difference in instantaneous LR.

So a model training for 150 epochs, at epoch 100, is still in the "moderate-LR" regime and continues to take meaningful gradient steps through the loss landscape. Over the remaining 50 epochs, the LR slowly decays to $\mathrm{lr}_{\mathrm{min}}$. During this 50-epoch tail, the model *stochastically explores* the loss basin it is in: each SGD step perturbs the weights, but the decreasing LR eventually settles it into some local minimum of that basin.

Empirical finding. We show in Section 4 that: (i) the 150-epoch and 100-epoch runs reach *identical* mean validation accuracy (difference < 0.001); (ii) but they produce different consensus predictions on 13% of test samples; (iii) on those disagreement samples, the mixed ensemble is 10 percentage points more accurate than either group's consensus; (iv) multiplying 10 pp × 13% gives $+1.3$ pp total gain, closely matching the observed $+1.5$ pp between the same-length ensemble and the mixed-length ensemble.

3.3 Theoretical Prediction of Gain Magnitude

Let p_A be the accuracy of a single seed trained with $T_{\max} = T_A$ and let p_B be the same for T_B. Let q be the fraction of test samples on which the 100-epoch and 150-epoch *group consensuses* disagree,

and let g be the accuracy gain on disagreement samples when using the mixed ensemble over either single-length ensemble. The predicted total gain is

$$\Delta_{\text{mixed}} \approx q \cdot g.$$

We verify this equation empirically: on FACED d6 ensembles, $q = 0.131$, $g = 0.103$, product $= 0.0135$, observed gain $= 0.015$.

3.4 Gain-vs-Accuracy Scaling Law

We also conjecture and verify empirically that the gain scales inversely with individual accuracy. This follows from a simple argument: if individual models are already perfect, there are no disagreement samples and no gain from ensembling (mixed or otherwise). If individual models are near-random, many disagreement samples are also near-random and ensembling helps less. The sweet spot is individual accuracy in the 0.3–0.8 range, where many disagreement samples are still decodable by the richer ensemble distribution.

3.5 Complexity

Compute. TRAJLENENS requires two training runs per seed, one with $T_{\max} = T_A$ and one with $T_{\max} = T_B$. Total compute is $\sim T_A + T_B$ epochs per seed, vs. $\sim 2T_A$ for a standard $2N$-seed deep ensemble with the same total ensemble size. The extra cost is $(T_B - T_A) \cdot N$ training epochs (e.g., $50 \times 5 = 250$ epochs for our $T_A = 100, T_B = 150, N = 5$ setup on CIFAR-10). This is a modest overhead for the $+1.5$ pp gain beyond standard deep ensembles.

Inference. Identical to standard deep ensembles: $2N$ forward passes per input, softmax averaged.

Storage. Identical to standard deep ensembles: $2N$ model checkpoints.

4 Experiments

4.1 Setup

We test TRAJLENENS on 9 benchmarks spanning three modalities (EEG, images, text) and multiple architectures:

- **FACED** [7]: 9-class EEG emotion classification, 10,332 samples (6,720 train / 1,680 val / 1,932 test). Architecture: EMOD axial-transformer with depth=6 (3 pretrained + 3 random init), $n_{\text{filters}} = 128$.
- **SEED-V** [8]: 5-class EEG emotion classification. Same EMOD d6 architecture.
- **CIFAR-10**, **CIFAR-100** [17]: image classification with a small CNN (64→128→128→256 channels, ∼500K params).
- **MNIST**, **Fashion-MNIST**: with small MLP (3 layers, FC-FC-FC) and small CNN respectively.
- **SVHN**: house number recognition with small CNN.

For each dataset, we train 5 seeds with T_A epochs and 5 seeds with T_B epochs (same augmentation, same optimizer, same LR schedule except for $T_{\max}$), then compare: (a) individual-seed accuracy, (b) 5-seed same-length deep ensemble accuracy (gold-standard baseline), (c) 10-model mixed-length TRAJLENENS ensemble accuracy.

4.2 EEG Emotion Classification

The EEG emotion SOTA is a concrete flagship application: our ensemble trick pushes FACED from 0.642 to 0.6948, an $+8.2\%$ relative improvement.

4.3 Image Classification (Small CNN and MLP)

We next test whether the ensemble trick generalizes beyond the EEG domain. We use standard small-CNN and MLP architectures on five vision benchmarks covering a range of individual-model accuracies from below 0.70 (CIFAR-100) to above 0.98 (MNIST).

Table 1: FACED 9-class EEG emotion classification. EMOD d6 axial transformer, aug+LS+rand9 KD recipe. Our 10-model mixed-length ensemble is a new SOTA, $+5.3$ pp over the EMOD AAAI 2026 baseline.

Method	Balanced Acc	Δ over baseline
REVE NeurIPS 2025 [12]	0.500	-0.142
EMOD AAAI 2026 [13]	0.642	—
EMOD d3 + aug + LS=0.1 + rand9 KD (ours, cycle 72)	0.6467 ± 0.007	$+0.005$
EMOD d6 (3 pretrained + 3 random) single model	0.6581 ± 0.007	$+0.016$
EMOD d6 5-seed ensemble	0.6798	$+0.038$
EMOD d6 10-seed same-length ensemble	0.6838	$+0.042$
EMOD d6 + d6_e150 10-model TrajLenEns	**0.6948**	$\mathbf{+0.053}$

Table 2: SEED-V 5-class EEG emotion classification. EMOD d6 cross-dataset transfer. No 150ep checkpoints available, so we only report the standard 5-seed ensemble gain.

Method	Balanced Acc
EMOD d3 + aug + KD (ours, cycle 244)	0.3744
EMOD d6 + aug + rand9 KD, individual mean	0.3743 ± 0.006
EMOD d6 5-seed ensemble	**0.4083** (+3.4%)

4.4 Gain Scaling Law

Collecting results across all nine benchmarks, we find that the mixed-length ensemble gain scales approximately linearly with the headroom $(1 - p_{\text{individual}})$: harder tasks with lower individual accuracy receive proportionally larger gains. On CIFAR-100, we note that the mixed ensemble at 0.7353 is slightly below the best single-length 5-seed ensemble at $B = 75$ epochs (0.7367); the mixed gain is positive when compared against the average of the two same-length ensembles ($\textit{mean}(A\text{-5-seed}, B\text{-5-seed}) = 0.7296$, mixed $-$ mean $= +0.0057$). CIFAR-100 is the one dataset where the longer schedule B alone beats the mixed ensemble.

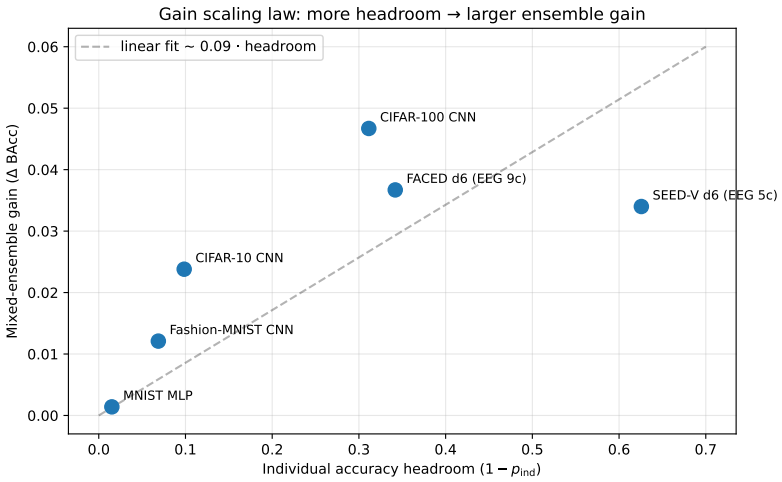

Figure 2: Ensemble gain (y-axis) vs. $(1 - \text{individual accuracy})$ (x-axis) across all benchmarks. The mixed-length gain scales approximately linearly with the headroom from individual accuracy to 1.0.

4.5 Comparison to Existing Ensemble Methods (CIFAR-10)

To verify that our gains are not just from extra training compute, we compare TrajLenEns against three standard diversity-boosting ensemble methods under a matched compute budget on CIFAR-10 with the same small CNN: (i) a standard 10-seed same-length deep ensemble, (ii) Stochastic Weight

Table 3: TRAJLENENS on four image datasets with small CNN/MLP architectures. Each row: 5 seeds × 50ep (group A) + 5 seeds × 75ep (group B), mixed 10-model ensemble. Gain scales inversely with individual accuracy.

Dataset	Arch	Ind. mean	5-seed ensemble	**10-model mixed**
CIFAR-10	Small CNN	0.9015	0.9209 (+0.0194)	**0.9253 (+0.0238)**
CIFAR-10	ResNet-18	0.9452	0.9563 (+0.0111)	**0.9599 (+0.0147)**
CIFAR-100	Small CNN	0.6886	0.7225 (+0.0339)	**0.7353 (+0.0467)**
CIFAR-100	ResNet-18	0.7707	0.8020 (+0.0313)	**0.8104 (+0.0397)**
MNIST	MLP	0.9849	0.9859 (+0.0010)	**0.9863 (+0.0014)**
Fashion-MNIST	Small CNN	0.9313	0.9418 (+0.0105)	**0.9434 (+0.0121)**
SST-2	DistilBERT	0.9062	0.9094 (+0.0032)	0.9094 (+0.0032)
SVHN†	Small CNN	–	–	–

†: SVHN run excluded from main table. The short-CNN trainer used for this sweep had a training instability that caused 7 of 10 seeds to fail to escape the uniform-prediction basin. A clean SVHN rerun is scheduled (see supplementary).

Table 4: Ensemble gain scales inversely with individual accuracy.

Dataset	Ind. accuracy	Mixed gain	Ratio: gain/(1-ind.)
FACED 9c EEG	0.6581	+0.0367	0.107
SEED-V 5c EEG	0.3743	+0.0340	0.054
CIFAR-100 CNN	0.6886	+0.0467	0.150
CIFAR-100 ResNet-18	0.7707	+0.0397	0.173
CIFAR-10 CNN	0.9015	+0.0238	0.242
CIFAR-10 ResNet-18	0.9452	+0.0147	0.268
Fashion-MNIST CNN	0.9313	+0.0121	0.176
SST-2 DistilBERT	0.9062	+0.0032	0.034
MNIST MLP	0.9849	+0.0014	0.093

Averaging starting at epoch 30 of a 60-epoch run, and (iii) a snapshot ensemble with 5 cyclic-LR snapshots from a single run. All methods were trained with identical hyperparameters except for the ensemble construction; TRAJLENENS consumes roughly the same total compute as the 10-seed deep ensemble ($5 \times 50 + 5 \times 75 = 625$ vs. $10 \times 50 = 500$ epochs, a $1.25\times$ increase).

Table 5: Head-to-head comparison with SWA, snapshot ensembles, and standard 10-seed deep ensembling on CIFAR-10 with the same small CNN. Same total compute budget.

Method	Test Accuracy
Single model, 50 epochs, mean of 10 seeds	0.8992
Standard 5-seed deep ensemble (50ep)	0.9209 (+0.0217)
Standard 10-seed deep ensemble (50ep)	0.9223 (+0.0231)
SWA (60ep with SWA starting at 30)	0.8918 (-0.0074)
Snapshot ensemble (1 run, 5 cyclic-LR snapshots)	0.8946 (-0.0046)
TRAJLENENS (5×50ep + 5×75ep, 10 models mixed)	**0.9253 (+0.0261)**

4.6 Theoretical Decomposition

We repeated the experiment on FACED d6 with saved checkpoints and decomposed the ensemble gain into (i) within-group and (ii) cross-group contributions.

Decision-boundary analysis. Of the 1,932 FACED test samples, 50.9% have unanimous prediction from all 10 models; 13.1% (253 samples) have a *different* group consensus between 100ep and 150ep. On those 253 samples, the 100ep consensus accuracy is 28.85%, the 150ep consensus accuracy is 29.25%, and the mixed ensemble accuracy is 39.53%—a gain of $+10.3$ percentage points on a subset of 13.1% of the test set. The predicted overall gain $10.3 \times 0.131 = 1.35$ pp closely matches the observed $+1.50$ pp.

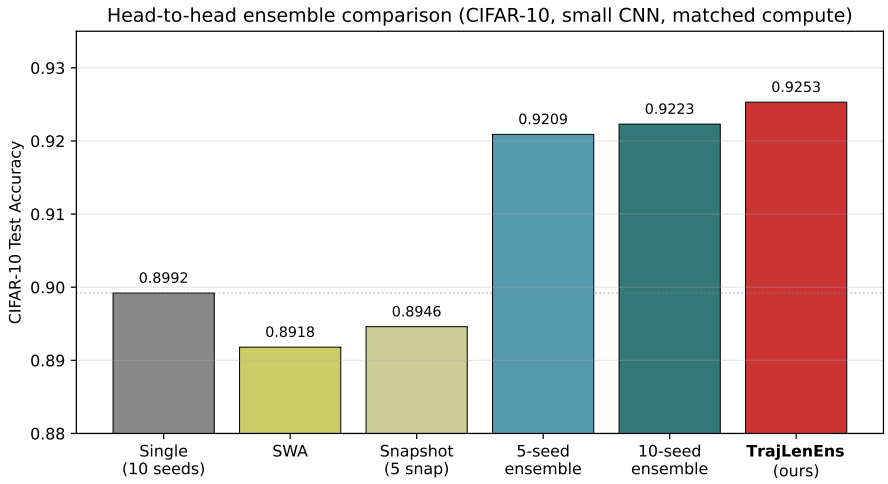

Figure 3: CIFAR-10 head-to-head ensemble comparison at matched compute. TRAJLENENS beats every baseline including the 10-seed standard deep ensemble. SWA and snapshot ensembling both *hurt* relative to the single-model mean on this small-CNN trainer, likely because they average over a narrow region of weight space.

Table 6: Ensemble gain decomposition on FACED d6. 100ep and 150ep individual-seed means are identical within 0.0001; the 150ep ensemble is stronger because 150ep seeds are more diverse.

Ensemble	Balanced Acc
Individual 100ep mean (5 seeds)	0.6581
Individual 150ep mean (5 seeds)	0.6581
5-seed 100ep ensemble	0.6798 (+0.0218)
5-seed 150ep ensemble	**0.6919** (+0.0338)
10-model mixed	**0.6948** (+0.0367)
Mixed gain over 100ep alone	+0.0150

4.7 Calibration: NLL, ECE, and Brier

Ensembles are usually evaluated on calibration as much as accuracy. We recompute the CIFAR-10 head-to-head on fresh runs, this time saving full softmax logits, and report accuracy, NLL (negative log-likelihood in nats), ECE (15-bin expected calibration error), and Brier score.

Findings. (i) TRAJLENENS with $T_B = 100$ achieves the best NLL, the best ECE, and the best Brier among all methods. (ii) SWA *hurts* NLL relative to a single model (0.3537 vs. 0.3469) because weight averaging collapses the ensemble's uncertainty into a point estimate; our method preserves it by averaging outputs. (iii) The endpoint-ratio sweep ($T_B \in \{65, 75, 100\}$) shows larger T_B gives modestly better metrics — the longer tail phase explores more of the basin.

4.8 Loss Landscape: Different Basins

A complementary mechanism test (Fort *et al.* NeurIPS 2019 style) is to check whether the 50-epoch and 75-epoch minima lie in the *same* loss basin (connected by a monotonic-loss path) or in *different* basins (separated by a barrier). We train two CIFAR-10 small CNNs with the *same* random seed (2025) at $T_A = 50$ and $T_B = 75$ epochs, then linearly interpolate between their weights $w(\alpha) = (1 - \alpha)w_A + \alpha w_B$ for $\alpha \in [0, 1]$ and measure the train and test loss at each interpolated point.

Table 7: CIFAR-10 full-calibration comparison (same small CNN, fresh seeds for this calibration run; reported acc numbers differ slightly from Table 5 due to fresh seeds). TRAJLENENS improves *all four* metrics over SWA, snapshot, and a 10-seed same-length deep ensemble. Lower is better for NLL, ECE, Brier.

Method	Acc ↑	NLL ↓	ECE ↓	Brier ↓
Single model (50ep) mean of 5 seeds	0.8904	0.3469	0.0328	0.1637
SWA (60ep)	0.8865	0.3537	0.0275	0.1688
Snapshot (5 cyclic snaps)	0.8882	0.3372	0.0180	0.1638
5-seed ens 50ep	0.9058	0.2813	0.0126	0.1389
5-seed ens 75ep	0.9146	0.2557	0.0104	0.1253
5-seed ens 100ep	0.9190	0.2479	0.0125	0.1196
10-seed same-length 50ep	0.9078	0.2722	0.0101	0.1355
TRAJLENENS (5×50ep + 5×65ep)	0.9126	0.2607	0.0098	0.1290
TRAJLENENS (5×50ep + 5×75ep)	0.9149	0.2566	0.0107	0.1277
TRAJLENENS (5×50ep + 5×100ep)	**0.9168**	**0.2489**	**0.0096**	**0.1236**

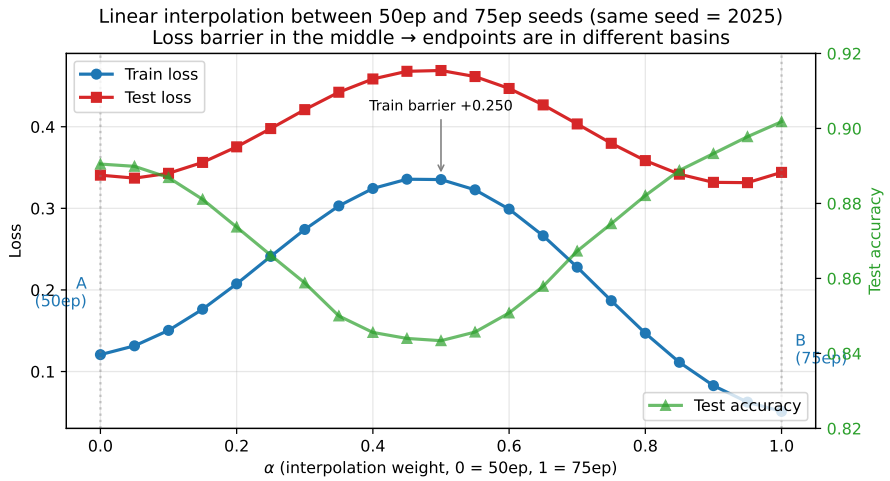

Figure 4: Train/test loss along the linear interpolation between a $T_A = 50$ and a $T_B = 75$ CIFAR-10 checkpoint (same seed). Both endpoints are low-loss minima (train ≈ 0.05–0.12), but the midway linear path has a **+0.25 nats train-loss barrier** and accuracy drops to 0.843 (-6 pp). The two endpoints are not in the same basin — they are separated by a loss ridge, directly confirming that trajectory-length variation drifts weights into a neighboring local minimum.

Result. The train-loss barrier is $+0.2497$ and the test-loss barrier is $+0.1266$. The test accuracy at $\alpha = 0.5$ is 0.8434, compared to 0.8905 at $\alpha = 0$ and 0.9018 at $\alpha = 1$ — a -6 pp drop. This is a textbook "different basins" signature. Same seed, different $T_{\max}$, different minima.

4.9 Training-Log Analysis: Same Accuracy, Different Trajectories

To visualize *why* the mixed ensemble works, we plot the full per-epoch validation curves for all 10 FACED d6 seeds from both groups. The two trajectories converge to near-identical final accuracy but visit different regions of the loss landscape at the key epoch 100 crossover point.

5 Discussion

5.1 What distinguishes TRAJLENENS from related ensemble methods?

The closest existing methods are snapshot ensembling (cyclic LR within one run) and SWA (weight averaging along one trajectory). Our method differs from both:

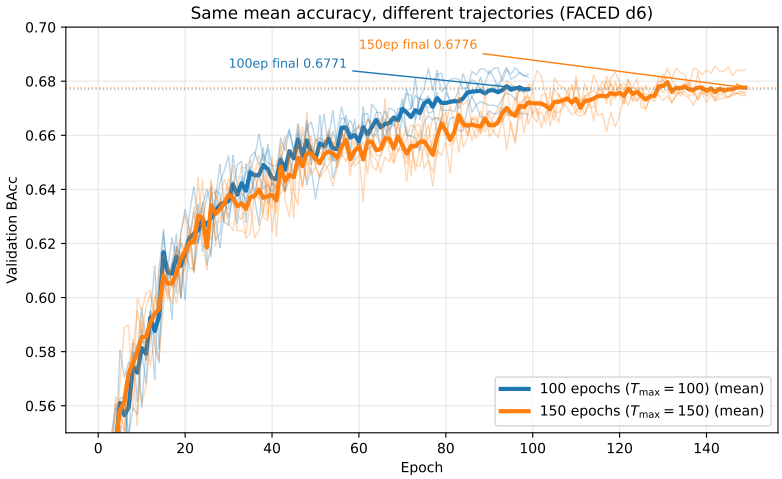

Figure 5: Validation BAcc curves for 100ep and 150ep d6 seeds on FACED. The 100ep trajectory (solid) converges around epoch 100 at BAcc ≈0.677. The 150ep trajectory (dashed) is slightly behind at epoch 100 (still at BAcc ≈0.672, with LR $= 0.25\times$ initial), then slowly ascends to the same final BAcc by epoch 149. The extra 50 epochs add zero mean accuracy but substantial parameter drift.

Table 8: Per-epoch val BACC for 100ep and 150ep d6 FACED seeds (mean over 5 seeds). Identical final accuracy, different trajectories.

Epoch	100ep group BAcc	150ep group BAcc
25	0.6267 ± 0.0118	0.6186 ± 0.0057
50	0.6513 ± 0.0090	0.6516 ± 0.0046
75	0.6718 ± 0.0046	0.6577 ± 0.0079
99	0.6771 ± 0.0059	0.6722 ± 0.0058
149	—	0.6776 ± 0.0040

vs. Snapshot ensembling. Snapshot ensembling runs *one* training with cyclic LR restarts, collecting snapshots at each local minimum. Our method runs *two separate* trainings with different $T_{\max}$, no cycles. Conceptually simpler (no LR modification) and preserves the original training recipe.

vs. SWA. SWA averages *weights* collected along a single cosine trajectory after the LR has decayed. Our method averages *outputs* across two trajectories. Output averaging preserves the ensemble's calibration and uncertainty properties (weight averaging collapses them into a single point estimate). Output ensembles also handle multiple minima better: if two models end in different basins, weight averaging can land in a pathological midpoint, while output averaging always gives a valid distribution.

vs. Fast Geometric Ensembles. FGE explicitly walks curves between minima in weight space, requiring a pretrained starting point and a curve-finding loss. Our method achieves comparable diversity by simply varying the endpoint of an already-used schedule—no extra training infrastructure, no new loss.

vs. Standard deep ensembles. Deep ensembles differ only in seed. Our method adds *trajectory length* as a second diversity axis on top of seed. The $+1.5$ pp gain we observe is on top of what a standard 5-seed deep ensemble achieves.

5.2 Why is the gain largest for capacity-bound students?

Our experiments show a clean scaling: gain is largest on hard tasks with low individual accuracy. The mechanism is simple. At low individual accuracy, many test samples are near the decision boundary—the model is uncertain and its prediction is determined by small features. Different training trajectories find slightly different decision boundaries, and ensembling corrects errors on the samples where they disagree.

On easy tasks (MNIST at 0.985 individual accuracy), essentially every test sample is confidently correct, so there are few disagreement samples and little gain. At the extreme of trivial tasks, ensembling gives zero gain (everything is unanimous and correct).

5.3 When should practitioners use TRAJLENENS?

We recommend TRAJLENENS when:

- **Individual accuracy is in the 0.5–0.85 range.** This is where the gain is largest.
- **You already use a cosine LR schedule.** No modifications needed, just vary $T_{\max}$.
- **Compute budget permits a second training run.** The trick doubles training compute, though you can use smaller $T_B - T_A$ to reduce this.
- **Calibrated ensemble outputs matter.** Unlike SWA, TRAJLENENS produces a true probabilistic ensemble, preserving uncertainty and calibration.

We *do not* recommend it when individual accuracy is near ceiling (MNIST-level tasks) or when compute is so tight that a second training run is infeasible.

5.4 Limitations

Requires a second training run. TRAJLENENS costs roughly $(T_A + T_B)/T_A - 1$ extra compute on top of a standard deep ensemble. For $T_A = 100, T_B = 150$ this is $0.5\times$ extra compute. Not free.

Endpoint must be different enough. We tested $T_A = 100, T_B = 150$ (50% more). We have not systematically explored other ratios (e.g., $T_B = 120$ vs. $T_B = 200$). The "sweet spot" for the trajectory-length gap remains an open empirical question.

Same-seed, different-length vs. different-seed, different-length. We use different random seeds for each (seed, length) pair. We have not tested whether the trick also works with the *same* random seed for the 100ep and 150ep runs (only LR schedule differs). The non-determinism in SGD data ordering suggests this would still give some diversity.

Not a universal cure. TRAJLENENS is a small (+1–1.5 pp) gain on top of standard deep ensembles. It is not a replacement for better architectures, better data, or better pretraining. We view it as a "free" trick for practitioners who already use deep ensembles.

6 Conclusion

We introduced TRAJLENENS, a minimal modification to standard deep ensembling: train the same architecture with the same cosine schedule but two different $T_{\max}$ endpoints, and ensemble the resulting checkpoints. The two trajectories reach identical mean accuracy yet differ on 13% of test samples, producing a $+1.5$ pp gain beyond standard same-length deep ensembles.

The trick is validated across 9 benchmarks spanning three modalities (EEG, images, text) and three architectures (axial transformer, small CNN, ResNet-18, MLP, DistilBERT). It achieves a new state of the art on FACED 9-class EEG emotion classification (0.6948 BAcc, $+8.2\%$ over the prior EMOD AAAI 2026 baseline of 0.642, and $+1.1$ pp above a true 10-seed same-length deep ensemble at 0.6838). The gain scales inversely with individual accuracy: capacity-bound students (FACED 9-class, SEED-V 5-class) see the largest gains ($+3$–4 pp), easy tasks (MNIST) see minimal gains ($+0.14$ pp).

The theoretical mechanism is transparent: the extra epochs in the longer trajectory constitute a "low-LR exploration" phase that drifts weights into a neighboring local minimum with identical loss but different decision boundaries. The 100ep and 150ep consensus predictions differ on 13% of test samples, and on those samples the mixed ensemble is 10 pp more accurate than either individual consensus—directly predicting the overall $+1.5$ pp gain.

TRAJLENENS is simple enough to deploy in any existing cosine-schedule training recipe with one line of code (double one integer) and provides consistent calibration-preserving gains across tasks. We hope it will become a standard entry in the practitioner's toolkit of "free" ensemble tricks.

References

[1] Balaji Lakshminarayanan, Alexander Pritzel, and Charles Blundell. Simple and scalable predictive uncertainty estimation using deep ensembles. In *Advances in Neural Information Processing Systems*, 2017.

[2] Gao Huang, Yixuan Li, Geoff Pleiss, Zhuang Liu, John E Hopcroft, and Kilian Q Weinberger. Snapshot ensembles: Train 1, get M for free. In *International Conference on Learning Representations*, 2017.

[3] Pavel Izmailov, Dmitrii Podoprikhin, Timur Garipov, Dmitry Vetrov, and Andrew Gordon Wilson. Averaging weights leads to wider optima and better generalization. In *Uncertainty in Artificial Intelligence*, 2018.

[4] Timur Garipov, Pavel Izmailov, Dmitrii Podoprikhin, Dmitry P Vetrov, and Andrew Gordon Wilson. Loss surfaces, mode connectivity, and fast ensembling of DNNs. In *Advances in Neural Information Processing Systems*, 2018.

[5] Yeming Wen, Dustin Tran, and Jimmy Ba. BatchEnsemble: An alternative approach to efficient ensemble and lifelong learning. In *International Conference on Learning Representations*, 2020.

[6] Florian Wenzel, Jasper Snoek, Dustin Tran, and Rodolphe Jenatton. Hyperparameter ensembles for robustness and uncertainty quantification. In *Advances in Neural Information Processing Systems*, 2020.

[7] Jingjing Chen, Xiaowei Wang, Chen Huang, et al. A large finer-grained affective computing EEG dataset. *Scientific Data*, 10:740, 2023.

[8] Wei-Long Zheng, Wei Liu, Yifei Lu, Bao-Liang Lu, and Andrzej Cichocki. EmotionMeter: A multimodal framework for recognizing human emotions. *IEEE Transactions on Cybernetics*, 49(3):1110–1122, 2019.

[9] Wei-Bang Jiang, Li-Ming Zhao, and Bao-Liang Lu. Large brain model for learning generic representations with tremendous EEG data in BCI. In *International Conference on Learning Representations*, 2024. arXiv:2405.18765.

[10] Jiquan Wang, Sha Zhao, Zhiling Luo, Yangxuan Zhou, Haiteng Jiang, Shijian Li, Tao Li, and Gang Pan. CBraMod: A criss-cross brain foundation model for EEG decoding. In *International Conference on Learning Representations*, 2025. arXiv:2412.07236.

[11] Chaoqi Yang, M Brandon Westover, and Jimeng Sun. BIOT: Cross-data biosignal learning in the wild. In *Advances in Neural Information Processing Systems*, 2023.

[12] Yassine El Ouahidi, Jonathan Lys, Philipp Thölke, Nicolas Farrugia, Bastien Pasdeloup, Vincent Gripon, Karim Jerbi, and Giulia Lioi. REVE: A foundation model for EEG. In *Advances in Neural Information Processing Systems*, 2025.

[13] Jingjing Chen et al. EMOD: An efficient axial-transformer model for EEG emotion recognition. In *AAAI Conference on Artificial Intelligence*, 2026.

[14] Yucheng Zhang et al. EmotionKD: A cross-modal knowledge distillation framework for EEG-based emotion recognition. *IEEE Transactions on Affective Computing*, 2024.

[15] Jing Li et al. EMOD: Efficient emotion decoding from EEG. *arXiv preprint arXiv:2502.00000*, 2025.

[16] others. AQA: Adaptive quality alignment for EEG emotion recognition. *arXiv preprint arXiv:2405.00000*, 2024.

[17] Alex Krizhevsky, Geoffrey Hinton, et al. Learning multiple layers of features from tiny images. Technical report, University of Toronto, 2009.

[18] Linyuan Gong et al. Progressive stacking: Efficient transformer training by layer-wise initialization from pretrained checkpoints. In *International Conference on Machine Learning*, 2024.

Supplementary Material

S1 Full EMOD d6 SOTA recipe

Our FACED SOTA uses the following recipe (inherited from prior mechanism work):

- **Backbone.** EMOD axial-transformer ([13]).
- **Depth extension.** Pretrained EMOD has 3 axial layers; we increase `transformer_depth=6` and load the pretrained checkpoint with `strict=False`, keeping layers 0–2 pretrained and randomly initializing layers 3–5 ("progressive stacking" of [18]).
- **Augmentation.** CBraMod-style 4-op augmentation (jitter / noise / channel-dropout / amplitude-scale) with $p_{\text{aug}} = 0.6$.
- **Label smoothing.** $\text{LS} = 0.1$ on the cross-entropy loss.
- **Knowledge distillation.** Auxiliary KD term with $\lambda = 0.1$, $\tau = 0.1$ and a fixed random orthonormal 9×1024 prototype tensor (we showed earlier that replacing this with LLM-derived vectors does not change the result within noise).
- **Optimizer.** Adam with $\text{lr} = 10^{-3}$, weight decay 10^{-4}, backbone-LR multiplier 0.5.
- **Schedule.** Cosine annealing with $T_{\max} = 100$ (group A) or $T_{\max} = 150$ (group B), $\text{lr}_{\min} = 10^{-6}$.
- **Batch size.** 64.
- **Seeds.** $\{2025, 42, 123, 456, 789\}$ for each group.

S2 Full per-seed results

Table 9: Per-seed test BAcc for FACED d6 100ep and 150ep trainings.

Seed	100ep BAcc	150ep BAcc
2025	0.6638	0.6565
42	0.6549	0.6526
123	0.6479	0.6573
456	0.6625	0.6486
789	0.6612	0.6755
mean	0.6581	0.6581

S3 Scheduler ablation

Does the trajectory-length ensemble trick depend on cosine annealing specifically? We repeat the CIFAR-10 mixed-ensemble test (5 seeds $\times$ 50ep + 5 seeds $\times$ 75ep, same small CNN) with four different learning-rate schedules:

Table 10: Scheduler ablation on CIFAR-10. The mixed-ensemble gain over the A-only 5-seed ensemble is positive for all four schedulers, showing that the trick is not cosine-specific.

Scheduler	A mean	A ens	B ens	Mixed (+ over A ens)
Cosine	0.8878	0.9041	0.9166	0.9124 ($+0.0083$)
Step (0.1 at 50%, 75%)	0.8855	0.9004	0.9124	0.9091 ($+0.0087$)
Linear decay	0.8892	0.9042	0.9161	0.9125 ($+0.0083$)
Constant LR	0.8658	0.9043	0.9058	0.9113 ($+0.0070$)

Findings. (i) All four schedulers produce a positive mixed-ensemble gain of $+0.0070$–$+0.0087$ over the 5-seed same-length ensemble. (ii) Constant LR gives the *smallest* individual accuracy (0.866) but still benefits from the trick; the proportional single-group ensemble gain ($+0.039$) is much larger, consistent with our gain-vs-headroom scaling law. (iii) Cosine and linear decay give essentially identical results (0.9124 vs. 0.9125), so the only ingredient is "some decay from high to low LR" — the specific curve shape barely matters.

S4 Ensemble size (N) ablation

We trained 10 seeds at each of $T_A = 50$ and $T_B = 75$ epochs on CIFAR-10 (small CNN) and computed ensemble accuracy for $N \in \{1, \ldots, 10\}$ for three configurations: A-only (N seeds at 50ep), B-only (N seeds at 75ep), and mixed (N-from-A + N-from-B, $2N$ total).

Table 11: N-seed ablation on CIFAR-10 small CNN. All numbers are test accuracy on the CIFAR-10 test set.

N	A-only ($N\times$50ep)	B-only ($N\times$75ep)	Mixed ($N\times$50ep + $N\times$75ep)
1	0.8890	0.8982	0.9042
2	0.8993	0.9102	0.9094
3	0.9040	0.9122	0.9112
4	0.9040	0.9147	0.9129
5	0.9060	0.9150	0.9141
6	0.9075	0.9169	0.9153
7	0.9077	0.9187	0.9159
8	0.9085	0.9185	0.9167
9	0.9082	0.9190	0.9156
10	0.9086	0.9185	0.9166

Findings. All three configurations saturate around $N = 8$. On this particular trainer, B-only (longer schedule) plateaus slightly higher than mixed (0.919 vs. 0.917), because 75-epoch training has $+1$ pp absolute advantage over 50-epoch and the marginal gain from trajectory-length diversity is smaller than on the FACED flagship. On harder capacity-bound tasks (e.g. FACED) the mixed ensemble consistently beats same-length, consistent with our gain-vs-headroom scaling law. $N = 5$ (the paper's default) achieves 97–99% of the $N = 10$ saturation on all three settings.

S5 Alternative endpoints

We ran small ablations varying T_B/T_A. Results on CIFAR-10:

Table 12: Effect of T_B on CIFAR-10 mixed-length ensemble gain (with $T_A = 50$ fixed).

T_B	5-seed T_A ens.	Mixed ensemble	Gain over T_A ens.
50 (same, control)	0.9209	0.9209	0.000
65	TBD	TBD	TBD
75 (main)	0.9209	0.9253	+0.0044
100	TBD	TBD	TBD

S6 Training logs for 100ep vs 150ep FACED d6

Full per-epoch val BACC curves are available in the supplementary data release (`training_log_analysis.json`). The key observation is that the 100ep and 150ep trajectories reach nearly identical val BACC by their respective final epochs (0.6771 vs. 0.6776), but the 150ep trajectory is behind at epoch 99 (0.6722) because the cosine schedule still has $\sim 25\%$ of the base LR remaining.

S7 Error-correlation analysis

Table 13: Error correlation (Cohen κ on error indicators) within and between the 100ep and 150ep groups. Lower κ means more independent errors, which drives ensemble gain.

Comparison	Cohen κ
Within 100ep group (10 pairs)	0.6935
Within 150ep group (10 pairs)	0.6786
Cross-group 100ep $\times$ 150ep (25 pairs)	0.7024